



Stash House

Reconstituting Disciplined Secret Use under Local Disorder

Franklin Diaz

University of Colorado

March 2026

TECHNOLOGY PRESENTATION

The Problem: Local Disorder

From Chaos to Stewardship

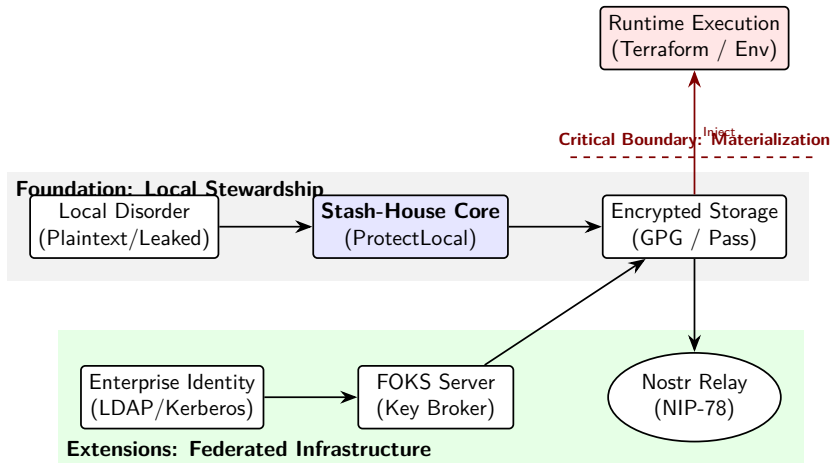
Admitting the Entropy Problem

- ▶ **Credential Sprawl:** Cloud tokens, SSH keys, and DB credentials in history files and unencrypted artifacts.
- ▶ **Passive vs. Active:** Detection (GitGuardian/Gitleaks) is not remediation.
- ▶ **The Goal:** Reconstitute discipline without destroying developer velocity.



Figure: The state of local disorder.

System Architecture

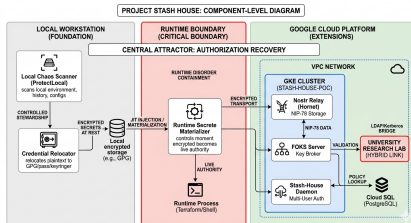


Authority Recovery

The Foundation

The ProtectLocal Operator

- ▶ **Scan-Identify-Relocate:** Automating the discovery of tokens in known and unknown artifacts.
- ▶ **Local Stewardship:** Utilizing GPG-backed pass hierarchies for immediate containment.
- ▶ **Hygiene Boundary:** Establishing a point of no return for plaintext secrets.



The Materialization Boundary

Securing the Injection Point

The Risk of Live Secrets

- ▶ **Defining the Boundary:** The precise moment encrypted authority becomes live runtime authority.
- ▶ **Temporal Exposure:** Minimizing the window between decryption and process consumption.
- ▶ **JIT Patterns:** Just-in-Time injection into shell environments without disk artifacts.

Federated Infrastructure

Sovereign Identity meets Enterprise Governance

Nostr Transport & NIP-78

- ▶ **Distributed Redundancy:** Moving from local-only to decentralized event-based storage.
- ▶ **NIP-78 (Application Data):** Virtual Vaults persisted across relay networks.
- ▶ **Cryptographic Sovereignty:** Data is signed and encrypted; relays are content-agnostic.

FOKS & Enterprise Bridging

- ▶ **FOKS (Federated Open Key Service):** Brokered key release via hardware tokens.
- ▶ **The Bridge:** Site-to-site tunnels to institutional LDAP and Kerberos servers.
- ▶ **Mapping Identity:** Projecting sovereign workstation identity into enterprise governance models.

Conclusion: Disciplined Identity

- ▶ **Recovery > Detection:** Active relocation of authority is the path to hygiene.
- ▶ **Containment:** Managing the materialization boundary reduces runtime risk.
- ▶ **Scale:** Decentralized protocols (Nostr/FOKS) enable sovereign security at enterprise scale.

Questions?

github.com/devsecfranklin/stash-house